

• Experiments:-

Study of LAN Topology, Network Devices and Address Configuration

Objective:

1. To study the LAN topology used in college campus.
2. To identify networking device and understand their function.
3. To analyse wired and wireless communication method used in the network.
4. To measure speed of network for wired and wireless connections.
5. To obtain MAC, IP address and subnet mask.

Problem Statement

1. Draw the Network Topology:
 - Identify and draw the LAN topology of your organization (e.g. star, Bus, mesh).
 - Clearly label all components such as Router, Switch, Access Point and end devices (PCs, Laptops etc.).

Solⁿ The College Campus network observed this study consists of

- * Layer-3 core switches in the server Room
- * Managed by POE+ Switches
- * Wireless Access point (APs)
- * Ethernet port in classroom & Laboratories

Network Topology

Hybrid Hierarchical topology which combines:

- 1) Tree Topology
- 2) Extended star Topology
- 3) Ring Topology (Server Room)

The network hierarchical Layer

- 1) Core Layer
- 2) Distribution Layer
- 3) Access Layer

1) Core Layer

- Core Layer contains server room and server room have Firewall, Router, 2-L3+ switches, Patch Rack, Servers, inverter etc.

2) Distribution Layer

- Distribution Layer contains 4-switches.
- The 4-switches are -

- 1) Main Build → R7 (Room no-213, 2nd Floor)
- 2) Main Build → R1 (Room no-301, 3rd Floor)
- 3) Main Build → R6 (Room no-113, 1st Floor)
- 4) Main Build → R2 (Room no-201, 2nd Floor)

⇒ This 4-switches contains

- 1) 18 Port Gigabit Easy smart-switch with 16-port. TL-SG1218MPE

- 2) SG3428XMP - 24-Port Gigabit & 4-port 10Gb SFPs L2+ managed switches with 24 Port PoE+.

→ Now let's discuss these 2 above components -

1) TP-Link TL-SG1218 MPE

- Likely used for: CCTV camera, wifi access point etc.
- 18 port Gigabit "Easy Smart" switch.
- 16 port support PoE+

2) TP-Link Omada SG3428 XMP

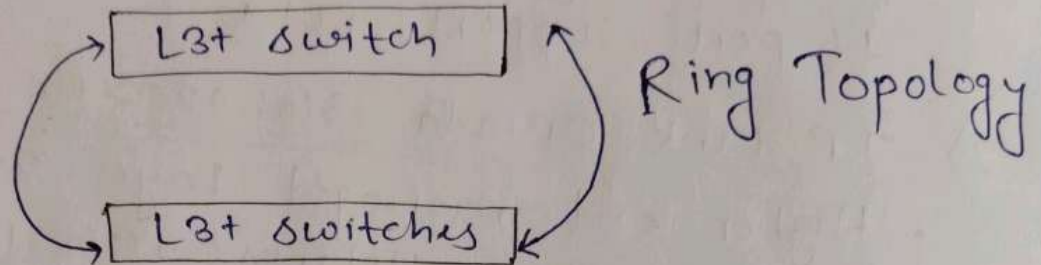
- Higher end managed Layer 2+ switch.
- 24 PoE+ Port, Uplinks, 10G SFP
- Usually acts as a main distribution for each floor

<3> Access Layer

- Access Layer contain 3- Switches
i.e.
 - 1) Main Build R1- (Room no-301, 3rd floor)
 - 2) Main Build R4- (Room no-05, Ground Floor)
 - 3) Main Build R5- (Room no-18, Ground Floor)
- Every Switches acts as access layer which helps to provide network connectivity to all class room, Lab Faculty Room etc.
- Access Layer directly connect:
 - 1) Ethernet Port
 - 2) Wireless AP
 - 3) Lab System
 - 4) Outdoor wireless AP (ex-D18)

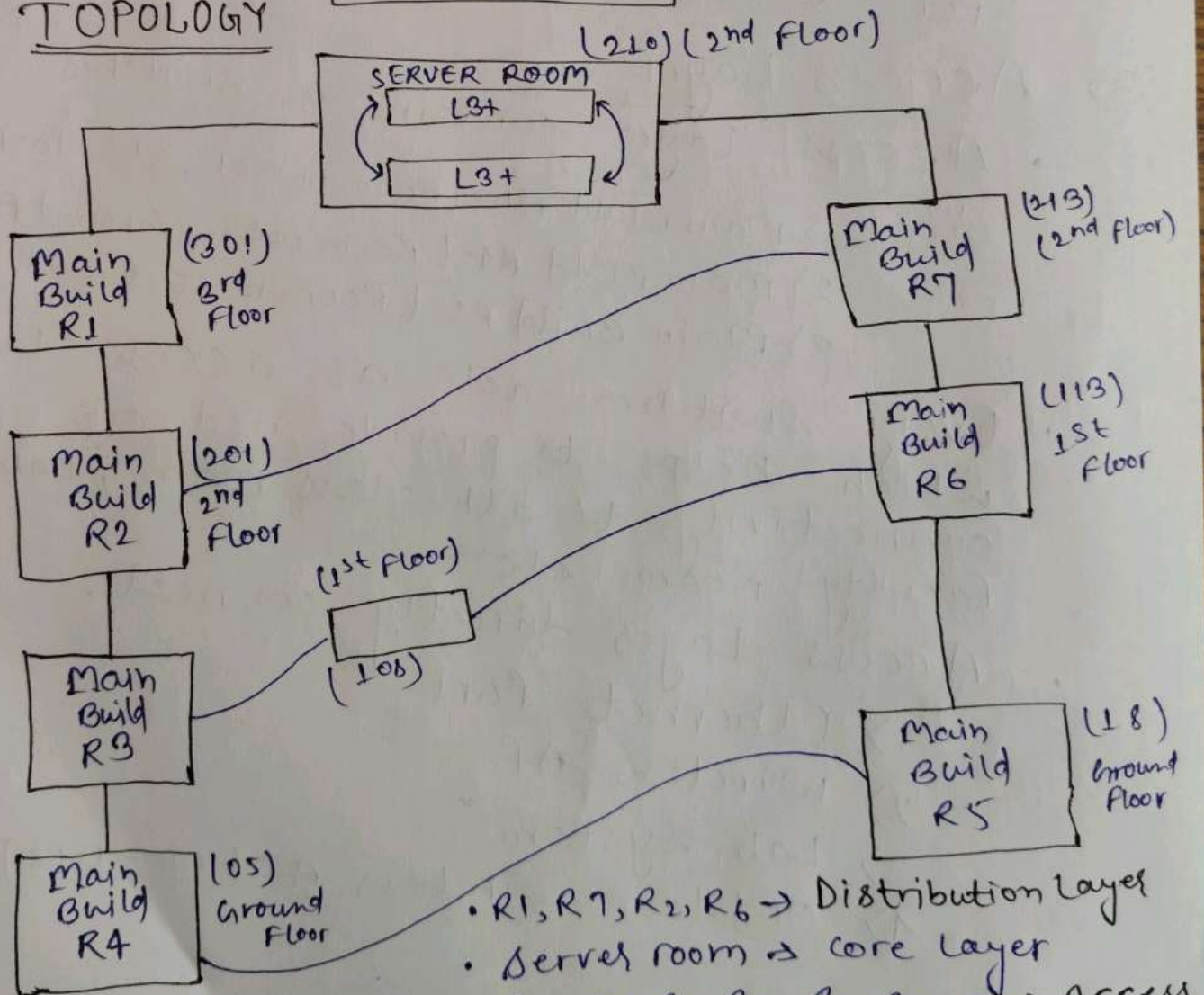
Server Room Architecture / Topology

ROUTER



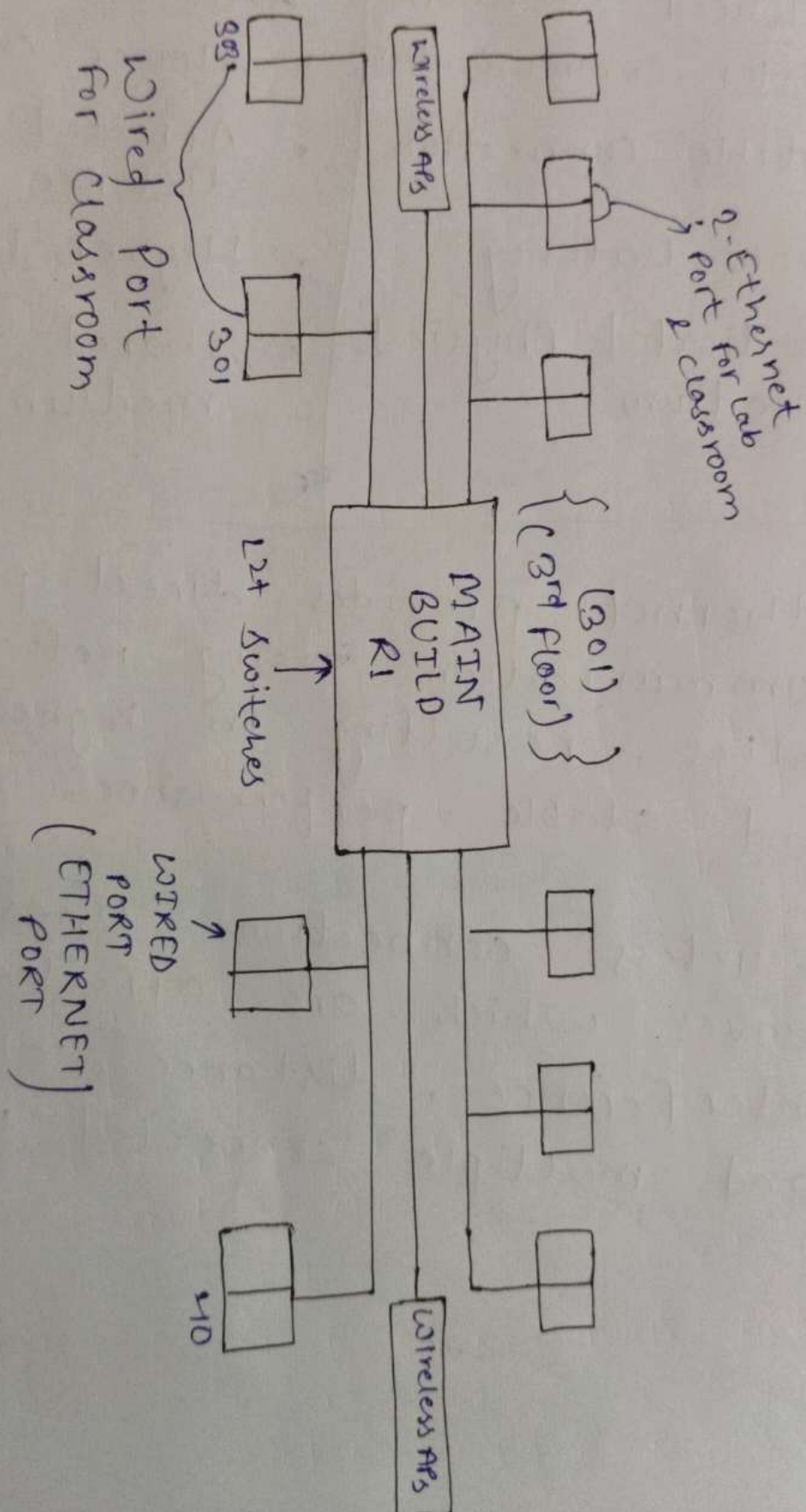
L2 Switches

• TOPOLOGY



- R1, R7, R2, R6 → Distribution Layer
- Server room → Core Layer
- R1, R2, R3, R4, R5, R6, R7 → Access Layer

ACCESS LAYER SWITCH (3rd Floor - 301)



Wireless Connection

```
samit-kumar@samit-hp-ubuntu: ~  
$ ifconfig  
enp1s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether c8:d3:ff:e3:e2:f1 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 2850 bytes 242661 (242.6 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2850 bytes 242661 (242.6 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.183.207.130 netmask 255.255.255.0 broadcast 10.183.207.255  
    inet6 fe80::1839:7f42:4ab1:317a prefixlen 64 scopeid 0x20<link>  
    inet6 2401:4900:b3db:938a:e3ad:91a7:cbae:f7f5 prefixlen 64 scopeid 0x0<global>  
    inet6 2401:4900:b3db:938a:bbde:f750:d8ed:ae1a prefixlen 64 scopeid 0x0<global>  
    ether d0:57:7b:cd:21:f2 txqueuelen 1000 (Ethernet)  
    RX packets 76080 bytes 50365170 (50.3 MB)  
    RX errors 0 dropped 672 overruns 0 frame 0  
    TX packets 22555 bytes 7938565 (7.9 MB)  
    TX errors 0 dropped 14 overruns 0 carrier 0 collisions 0  
  
samit-kumar@samit-hp-ubuntu:~$
```

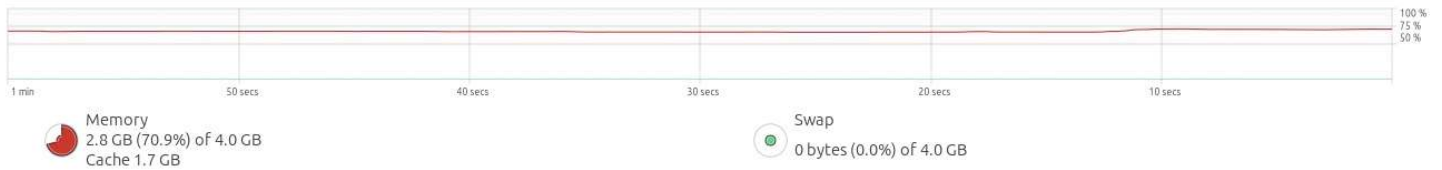
MAC Address : d0:57:7b:cd:21:f2
IP Address(IPv4) : 10.183.207.130
Subnet Mask : 255.255.255.0

Processes Resources File Systems

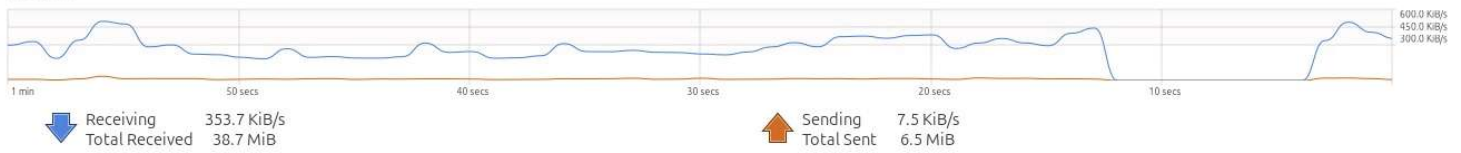
CPU



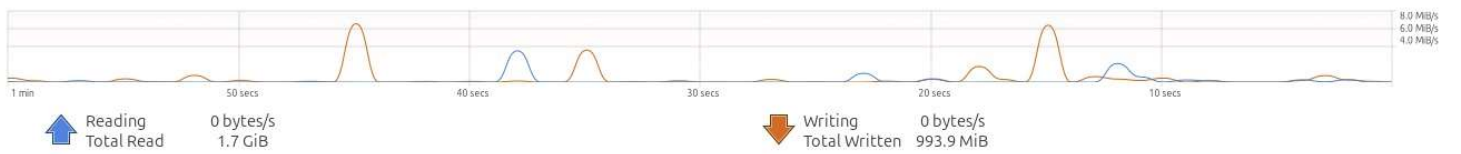
Memory and Swap



Network



Disk



Wired Connection

```
samit-kumar@samit-hp-ubuntu: ~  
$ ifconfig  
enp1s0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500  
    ether c8:d3:ff:e3:e2:f1 txqueuelen 1000 (Ethernet)  
    RX packets 0 bytes 0 (0.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 0 bytes 0 (0.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 2900 bytes 246972 (246.9 KB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 2900 bytes 246972 (246.9 KB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
wlp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 10.183.207.130 netmask 255.255.255.0 broadcast 10.183.207.255  
    inet6 fe80::1839:7f42:4ab1:317a prefixlen 64 scopeid 0x20<link>  
    inet6 2401:4900:b3db:938a:bbde:f750:d8ed:ae1a prefixlen 64 scopeid 0x0<global>  
    inet6 2401:4900:b3df:aadc:2cb3:55ac:41c5:347e prefixlen 64 scopeid 0x0<global>  
    inet6 2401:4900:b3df:aadc:b719:3d60:83b4:fc6a prefixlen 64 scopeid 0x0<global>  
    ether d0:57:7b:cd:21:f2 txqueuelen 1000 (Ethernet)  
    RX packets 98359 bytes 79092463 (79.0 MB)  
    RX errors 0 dropped 672 overruns 0 frame 0  
    TX packets 32581 bytes 9207167 (9.2 MB)  
    TX errors 0 dropped 14 overruns 0 carrier 0 collisions 0  
  
samit-kumar@samit-hp-ubuntu:~$
```

MAC Address :d0:57:7b:cd:21:f2
IP Address(IPOv4) : 10.183.207.130
Subnet Mask: 255.255.255.0

Difference between wired and wireless speed.

Wired Network

- Faster Communication
- Stable Connection
- Low Latency
- Dedicated Physical medium

Wireless Network

- * Slower Communication
- Affected by distance / interface
- Higher Latency
- Shared radio medium

- Ethernet provides direct physical communication using network cables, resulting in higher speed and stable performance.
- Wireless connections use radio waves which are affected by interference, distance, obstacles and multiple connected users.

Q. Role of subnet mask in network

Defⁿ . A Subnet mask identifies the network portion and host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

It is important for:

- 1) Network Segment
- 2) Efficient Routing
- 3) Communication between devices.

OBSERVATION

During the study of the college network infrastructure

1. A centralized server room was used as network backbone.
2. Two L3 switches were connected in redundant cyclic structure.
3. Managed TP-Link PoE+ switches were used in network cabinets.

* Advantage of Existing Network

- Scalable architecture
- Centralized management
- Supports PoE devices
- Easy expansion of floor and classroom
- Support both wireless and wired
- Redundant link for failover.

CONCLUSION

The observed campus LAN follow a hybrid hierarchical topology consist of tree, Extended star, Ring.

The network used managed PoE switches, wireless access point Ethernet port and centralized server infrastructure to provide efficient wired and wireless communication. across the campus.